

In one line: Layout is a **negotiation**, not a constraint solver: the **parent proposes** a size, the **child chooses** its own size (it may ignore the offer), the **parent places** the child. Every modifier is a new parent in that chain — so **order matters**.

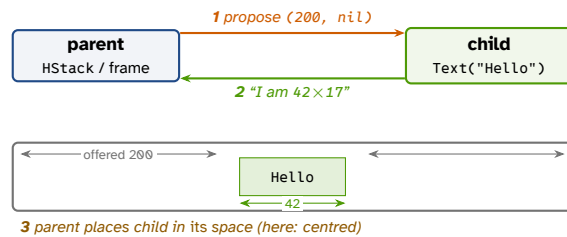
How it works

- **Three steps**, top-down proposal then bottom-up answer: (1) parent proposes a `ProposedViewSize`; (2) child returns a `CGSize` — its decision, never forced; (3) parent positions the child in its own coordinate space. One pass, no Auto Layout solver.
- **ProposedViewSize** — each axis is `CGFloat?`: `nil` = “your ideal” (`.unspecified`); `0` = “your minimum” (`.zero`); `.infinity` = “your maximum” (`.infinity`); a number = a real offer.
- **Fixed vs flexible:** `Text/Image` hug content (`Text` wraps then truncates when offered less); `Color`, `Shape`, `Spacer`, `GeometryReader` are greedy — they take whatever is offered.
- `.frame(width: 100)` proposes exactly 100 and reports 100, whatever the child does (it may overflow). `.frame(maxWidth: .infinity)` is a negotiating frame: it clamps the offer to `[min, max]` — fills *what the parent offered*, not the screen.
- `fixedSize()` replaces the incoming proposal with `nil` → child takes its ideal size. `fixedSize(horizontal: false, vertical: true)` = wrap, never truncate.
- **Stack distribution** (`HStack`): subtract spacing; probe each child's min/max to get its *flexibility*; highest `layoutPriority` group first (lower groups reserved at their min); within a group, least flexible child first, each offered *remaining ÷ children left* (*unverified*). Default priority 0.
- **Alignment:** the stack's alignment: picks the guide all children share (`.leading`, `.firstTextBaseline`); `.alignmentGuide(_:computeValue:)` lets *one* child move its own guide; a custom `AlignmentID` aligns views across containers.
- **GeometryReader** takes *all* proposed space, proposes it to its child and places the child **top-leading** (iOS 14+). Prefer `containerRelativeFrame(17)`, `onGeometryChange` or `.background { GeometryReader... }` to *read* size.
- **ViewThatFits** (16): tries children in order, shows the first whose *ideal* size fits the proposal (in: `.horizontal` to limit the axis).
- **Layout protocol** (16):
`sizeThatFits(proposal:subviews:cache:)` +
`placeSubviews(in:proposal:subviews:cache:)`; measure with `subview.sizeThatFits(_)`, place with `subview.place(at:anchor:proposal:)`.

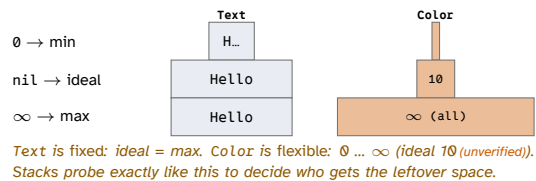
Example — a flow (wrap) layout

```
struct Flow: Layout {
    func sizeThatFits(proposal: ProposedViewSize, subviews: Subviews,
        cache: inout ()) -> CGSize {
        let w = proposal.replacingUnspecifiedDimensions().width
        var x = 0.0, y = 0.0, row = 0.0
        for s in subviews {
            let d = s.sizeThatFits(.unspecified) // ideal size
            if x + d.width > w { x = 0; y += row; row = 0 }
            x += d.width; row = max(row, d.height)
        }
        return CGSize(width: w, height: y + row)
    }
    func placeSubviews(in b: CGRect, proposal: ProposedViewSize,
        subviews: Subviews, cache: inout ()) {
        // same walk; s.place(at: CGPoint(x: b.minX+x, y: b.minY+y))
    }
}
```

One negotiation, three probes



Probing a view with proposals



Modifier order — each modifier wraps the view below it, so it is a new *parent*:

```
Text("A").padding().background(.red) // red incl. padding
Text("A").background(.red).padding() // red behind text only
Text("A").frame(width: 80).border(.blue) // border = 80pt box
Text("A").border(.blue).frame(width: 80) // border hugs the text
```

Traps

- ! “nil proposal means zero / fill” — it means **ideal**.
- ! `maxWidth: .infinity` not filling — an ancestor (`fixedSize`, an overlay, a greedy sibling) never *offered* the width. Spacers cannot create space either.
- ! `GeometryReader` around a small view balloons it and pins content top-leading; one per `List` row costs a layout pass each.
- ! `Color.red.frame(height: 200).ignoresSafeArea()` — the frame, not the colour, touches the edge; use `.background(Color.red.ignoresSafeArea())`.
- ! Layout cache: the default `updateCache` rebuilds it whenever the layout or a subview changes — it pays only with an incremental `updateCache`; keep `()` unless measured.
- ! `placeSubviews` points are in the *parent's* space — offset by `bounds.minX/minY`, never assume the origin is `(0,0)`.

Remember

“Parent proposes, child disposes, parent places.” 0 / nil / ∞ = min / ideal / max. Read modifiers *bottom-up* as parents wrapping children.